Chapter 8 — Chemical Equilibrium

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

Practice MCQs

1. At dynamic equilibrium:

(a) Both rates zero **(b)** Only reverse occurs **(c)** Forward rate = reverse rate **(d)** Only forward occurs

2. Equilibrium requires:

(a) Zero temperature **(b)** No catalyst **(c)** Open beaker always **(d)** Closed system

3. Kc expression uses:

(a) Atomic numbers **(b)** Volumes only **(c)** Initial masses **(d)** Equilibrium concentrations

4. Pure solids in K expression are:

(a) Omitted **(b)** Squared **(c)** Inverted **(d)** Included

5. Large K means:

(a) No reaction **(b)** Rate zero **(c)** Reactants favoured **(d)** Products favoured

6. K changes with:

(a) Concentration **(b)** Catalyst **(c)** Temperature **(d)** Pressure

7. Catalyst at equilibrium:

(a) Speeds both directions **(b)** Shifts right **(c)** Changes K **(d)** Shifts left

8. Le Chatelier principle opposes:

(a) Mass only **(b)** Catalyst only **(c)** Temperature only **(d)** Disturbance

9. Increasing pressure favours:

(a) Solids only **(b)** No side **(c)** More gas moles **(d)** Fewer gas moles

10. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, high pressure:

(a) Increases NH_3 **(b)** Stops reaction **(c)** Decreases NH_3 **(d)** No effect

11. Exothermic forward reaction, higher T:

(a) Favours forward **(b)** Favours reverse **(c)** Changes catalyst **(d)** No effect

12. Removing product shifts equilibrium:

(a) No shift **(b)** Reverse **(c)** Stops **(d)** Forward

13. Common ion effect:

(a) Changes Ksp **(b)** Decreases solubility **(c)** Increases solubility **(d)** No effect

14. Ksp of AgCl:

(a) [AgCl] **(b)** $[Cl^-]/[Ag^+]$ **(c)** $[Ag^+]/[Cl^-]$ **(d)** $[Ag^+][Cl^-]$

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15. Precipitation occurs when ionic product:

(a) Is always one **(b)** Exceeds Ksp **(c)** Equals zero **(d)** Is below Ksp

16. Kp uses:

(a) Partial pressures **(b)** Molar masses **(c)** Volumes **(d)** Masses

17. Haber process catalyst:

(a) Iron **(b)** Ni only **(c)** Pt only **(d)** V_2O_5

18. Contact process catalyst:

(a) Zn **(b)** Fe **(c)** V_2O_5 **(d)** Cu

19. Haber process temperature about:

(a) 100 °C **(b)** 1000 °C **(c)** 450 °C **(d)** 0 °C

20. Contact process forms:

(a) HCl **(b)** NH_3 **(c)** SO_3 **(d)** CH_4

21. Equilibrium concentrations are:

(a) Zero **(b)** Changing **(c)** Constant not equal **(d)** Always equal

22. Adding reactant shifts:

(a) Stops **(b)** Reverse **(c)** No shift **(d)** Forward

23. K for reverse reaction is:

(a) Zero **(b)** Same **(c)** Double **(d)** Reciprocal

24. If equation multiplied by 2, K becomes:

(a) Squared **(b)** Same **(c)** Halved **(d)** Doubled

25. Solubility product is constant at:

(a) Any pressure **(b)** Any concentration **(c)** Fixed temperature **(d)** Any catalyst

26. Adding NaCl to saturated AgCl:

(a) Forms NaAg **(b)** Dissolves more **(c)** No effect **(d)** Precipitates AgCl

27. Equilibrium can be reached from:

(a) Either direction **(b)** Only catalyst **(c)** Only forward **(d)** Only reverse

28. Macroscopic property at equilibrium:

(a) Random **(b)** Always changes **(c)** Remains constant **(d)** Zero

29. Microscopic events at equilibrium:

(a) Continue **(b)** Only reverse **(c)** Stop **(d)** Only forward

30. Industrial yield is improved by:

(a) Removing product **(b)** Adding catalyst only **(c)** Lowering K **(d)** Opening system

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31. Reversible reaction proceeds:

(a) Both directions **(b)** Forward only **(c)** Reverse only **(d)** No direction

32. Equilibrium is established in:

(a) Closed system **(b)** Open system always **(c)** No container **(d)** Vacuum only

33. Dynamic equilibrium means:

(a) Only forward occurs **(b)** Reactions stop **(c)** Rates equal **(d)** Concentrations zero

34. Macroscopic properties at equilibrium:

(a) Always change **(b)** Remain constant **(c)** Are random **(d)** Are zero

35. Microscopic events at equilibrium:

(a) Stop **(b)** Only reverse **(c)** Continue **(d)** Only forward

36. Kc uses:

(a) Volumes only **(b)** Equilibrium molar concentrations **(c)** Atomic numbers **(d)** Initial masses

37. Kp uses:

(a) Volumes only **(b)** Partial pressures **(c)** Densities **(d)** Molar masses

38. Pure liquids in K expression are:

(a) Squared **(b)** Omitted **(c)** Inverted **(d)** Added

39. Large K means:

(a) Reactants favoured **(b)** Products favoured **(c)** No reaction **(d)** Rate zero

40. Small K means:

(a) Reactants favoured **(b)** Products favoured **(c)** No reaction **(d)** Rate infinite

41. K for reverse reaction is:

(a) Zero **(b)** Reciprocal **(c)** Same **(d)** Double

42. If equation multiplied by 2, K becomes:

(a) Doubled **(b)** Halved **(c)** Same **(d)** Squared

43. Le Chatelier principle says system opposes:

(a) Catalyst only **(b)** Temperature only **(c)** Disturbance **(d)** Mass only

44. Adding reactant shifts equilibrium:

(a) Stops **(b)** No shift **(c)** Forward **(d)** Reverse

45. Increasing pressure favours:

(a) Solids only **(b)** More gas moles **(c)** Fewer gas moles **(d)** No side

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46. Increasing temperature favours:

(a) Endothermic direction **(b)** Exothermic direction **(c)** No direction **(d)** Catalyst only

47. Catalyst changes:

(a) Time to reach equilibrium **(b)** K **(c)** ΔH **(d)** Equilibrium yield

48. Homogeneous equilibrium has:

(a) Different phases **(b)** Same phase **(c)** Only solids **(d)** Only liquids

49. Heterogeneous equilibrium has:

(a) Different phases **(b)** No solids **(c)** Same phase only **(d)** Only gases

50. Hydrolysis of $BiCl_3$ is affected by:

(a) Density only **(b)** Molar mass only **(c)** Concentration of acid **(d)** Colour only

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Answer Key with Brief Reasons

1. Answer: (c) Forward rate = reverse rate — Dynamic balance.

2. Answer: (d) Closed system — No loss of matter.

3. Answer: (d) Equilibrium concentrations — Products over reactants.

4. Answer: (a) Omitted — Constant activity.

5. Answer: (d) Products favoured — Extent of reaction.

6. Answer: (c) Temperature — Only T changes K.

7. Answer: (a) Speeds both directions — No position change.

8. Answer: (d) Disturbance — System counteracts change.

9. Answer: (d) Fewer gas moles — Pressure effect.

10. Answer: (a) Increases NH_3 — 4 mol to 2 mol.

11. Answer: (b) Favours reverse — Heat as product.

12. Answer: (d) Forward — Replaces product.

13. Answer: (b) Decreases solubility — Suppresses ionization.

14. Answer: (d) $[Ag^+][Cl^-]$ — Sparingly soluble salt.

15. Answer: (b) Exceeds Ksp — Supersaturation.

16. Answer: (a) Partial pressures — Gas equilibrium.

17. Answer: (a) Iron — Book conditions.

18. Answer: (c) V_2O_5 — SO_2 oxidation.

19. Answer: (c) 450 °C — Compromise rate/yield.

20. Answer: (c) SO_3 — $SO_2 + O_2$.

21. Answer: (c) Constant not equal — Rates equal.

22. Answer: (d) Forward — Consumes added reactant.

23. Answer: (d) Reciprocal — Inverted expression.

24. Answer: (a) Squared — Powers multiply.

25. Answer: (c) Fixed temperature — Temperature dependent.

26. Answer: (d) Precipitates AgCl — Common Cl^-.

27. Answer: (a) Either direction — Same K.

28. Answer: (c) Remains constant — Observable constancy.

29. Answer: (a) Continue — Dynamic.

30. Answer: (a) Removing product — Shift forward.

31. Answer: (a) Both directions — Forward and reverse.

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32. Answer: (a) Closed system — No loss of matter.

33. Answer: (c) Rates equal — Continuous activity.

34. Answer: (b) Remain constant — Observable constancy.

35. Answer: (c) Continue — Molecular activity continues.

36. Answer: (b) Equilibrium molar concentrations — Products over reactants.

37. Answer: (b) Partial pressures — Gas equilibrium.

38. Answer: (b) Omitted — Constant activity.

39. Answer: (b) Products favoured — Extent of reaction.

40. Answer: (a) Reactants favoured — Extent of reaction.

41. Answer: (b) Reciprocal — Inverted expression.

42. Answer: (d) Squared — Powers multiply.

43. Answer: (c) Disturbance — Counteracts change.

44. Answer: (c) Forward — Consumes added reactant.

45. Answer: (c) Fewer gas moles — Pressure effect.

46. Answer: (a) Endothermic direction — Heat as reactant/product.

47. Answer: (a) Time to reach equilibrium — Speeds both directions.

48. Answer: (b) Same phase — One phase.

49. Answer: (a) Different phases — More than one phase.

50. Answer: (c) Concentration of acid — Le Chatelier application.

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